

Timber Constructability Audit

Sled runners and ramp planking under the published ramp-and-sled method. A forensic material-availability review of the conventional Great Pyramid construction method, with inputs taken from mainstream published Egyptology and the 2026 ramp-and-sled paper.

EDITION	Reader's Edition (free)
COMPANION TO	Quantifying Antiquity — Episode 4: Timber
WORKBOOK VERSION	v5.3
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FULL WORKBOOK	Faithful Export

This document presents the calculation, sources, and exclusions. All math is shown. Readers are invited to verify it. If the arithmetic is wrong, the document is wrong — that is why the math is on the table.

PREAMBLE

What this is and what it tests

This is a constructability audit of one specific element of the conventional Great Pyramid construction model — the structural timber required to operate the ramp-and-sled hauling method described in mainstream published Egyptology, including the 2026 peer-reviewed paper that quantifies haul distances and ramp parameters. The audit applies standard construction project-management methodology: a material takeoff against documented supply.

The calculation tests whether the documented cedar import record from the Old Kingdom — the Palermo Stone entry of forty ships under Snefru, the strongest import figure we have from the period — is sufficient to supply the structural timber the published method itself commits to.

WHAT IT DOES NOT CLAIM

This document does not claim alternative authorship, alien intervention, or lost civilizations. It does not dispute that the pyramids were built by ancient Egyptians, and it proposes no replacement method. It is a narrow, falsifiable arithmetic test of one method's implied material demand against one period's documented supply.

METHODOLOGY

All inputs are taken from mainstream published Egyptology or from standard construction engineering practice, and each is classified as **sourced**, **derived**, or **assumption**. The headline result is reported at a 100 km runner service-life assumption, which sits well toward the generous end of the sensitivity range. This edition reports two findings: the anchored runner-and-strip core, and — new in v5.3 — a fracture-ceiling extension that counts the sled itself as a consumable. The core is the defensible spine; the ceiling is documented and clearly flagged.

1. SCOPE OF ANALYSIS

Two timber components, conservatively scoped

WORKBOOK SHEET: README

The conventional ramp-and-sled method requires structural timber in at least six categories: sled runners and crossbars, ramp planking and sleepers, levers for final placement, scaffolding that grows with the pyramid, causeway and dock infrastructure at the Nile landing, and cribbing and block cradles during precision setting. The 2026 published paper explicitly requires sledges and ramp planking as functional components.

The **headline** models only the two components most directly exposed to wear:

In scope	Detail
Sled runners only	Two runners per sled. No crossbars, tow beams, deck, or lash points.
Ramp track-strip planking only	The two strips under runner contact. Not full ramp width.

Everything else in the method’s own timber requirements is excluded from the headline. The fracture-ceiling section (Section 6) then counts the sled body as a consumable and the omitted ledger (Section 7) catalogs the rest.

2. KEY INPUTS

The values that drive the calculation

WORKBOOK SHEET: Inputs

Input	Value	Unit	Source class
Average block dimensions	$1.27 \times 1.27 \times 0.71$	m	SOURCED
Limestone density	2,600	kg/m ³	SOURCED RANGE
Calculated block mass	2.98	tonnes	DERIVED
Sizing block mass	3.0	tonnes	DERIVED / CONSERVATIVE
Load + empty-return factor	1.20	factor	DERIVED
Total loaded haul (runner demand)	2,076,030	block-km	SESSION
Ramp haul (2026 paper)	696,030	block-km	SOURCED
Quarry-to-ramp haul (600 m avg)	1,380,000	block-km	DERIVED
Active ramps / length / lane width	4 / 580 / 3.8	—	SOURCED
Planked share of ramp surface	30	%	ASSUMPTION
Cedar density	510	kg/m ³	SOURCED
Cargo per cedar ship	35	tonnes/ship	ASSUMPTION
Documented cedar ships (Palermo / Snefru)	40	ships	SOURCED
Documented cedar benchmark	2,745	m³	DERIVED

3. RUNNER ENGINEERING BASIS

Sizing the soles of the sled

WORKBOOK SHEET: Runner_Calcs

The runner is sized to carry block plus sled at a contact pressure appropriate for cedar on a wet-sand or planked surface. The minimum structural thickness is the thickness below which the runner cannot safely carry the load. The new (as-installed) thickness adds a 50% sacrificial allowance — material worn off by friction before the runner reaches the minimum and must be replaced.

Parameter	Value	Basis
Runners per sled	2	Standard two-runner sled
Runner length × width	2.0 × 0.25	Sized for ~3 t block + tow geometry
Minimum structural thickness	0.15 m	6 in minimum for load
New runner thickness (min × 1.5)	0.225 m	50% sacrificial allowance
Allowable wear depth	75 mm	New – minimum
Runner pair replacement volume	0.225 m³	Cedar per replacement set
Block + sled loaded mass	3.35 t	Ground pressure ≈ 32.9 kPa

REPLACEMENT LOGIC

Wear rate determines replacement frequency. Cedar demand is based on the full replacement runner-pair volume, not the volume physically abraded away — standard construction-estimating practice. The 100 km service life is a sensitivity input, not a measured field life; the full table runs 50 km to 1,000 km in the Faithful Export. The wear rate itself is anchored to ASTM D2394-type wood-abrasion testing.

4. RAMP PLANK ENGINEERING BASIS

The floor the sled rides on

WORKBOOK SHEET: Ramp_Wear

Ramp planking is treated the same way as runners: a consumable wear surface, not a one-time installation. Each pass wears the plank by a fraction of the runner's own wear, because contact is wood-on-wood rather than wood-on-bedrock. When cumulative wear reaches the allowable depth, the plank is replaced.

Parameter	Value	Basis
New plank thickness (min × 1.5)	0.09 m	Allowable wear depth 30 mm
Active ramps × length	4 × 580 m	From 2026 paper
Full-width planked area	2,645 m ²	At 30% planked share
Track-strip area (runners only)	348 m ²	Two strips at 0.25 m
Plank wear-rate fraction	0.5	Plank wears at half runner rate

Two replacement cases are modeled. The headline uses the **track-strip** case — only planking under runner contact is replaced — because it is the minimum-replacement, most conservative interpretation. Full-width replacement is shown as an upper bound in the Faithful Export.

5. HEADLINE RESULT — 100 KM CASE

The anchored finding

WORKBOOK SHEET: Combined

Quantity	Value	Source
Total loaded haul	2,076,030 block-km	Ramp + quarry leg
Runner service life (assumed)	100 km	Sensitivity input
Runner sets required	20,760	Haul ÷ life × load factor
Cedar — runners only	5,605 m³	Sets × 0.225 × 1.20
Cedar — ramp track-strip planks	282 m³	Wear cycles × inventory
Combined cedar demand	5,887 m³	Runner + plank
Documented cedar benchmark	2,745 m³	40 ships × 35 t ÷ 510 kg/m³
Shortfall	3,142 m³	≈ 45.8 extra ships

2.14×

DEMAND VS DOCUMENTED
BENCHMARK

< half

OF THE METHOD'S NEED IS
COVERED

2 of 6

TIMBER CATEGORIES COUNTED

FINDING — 100 KM CASE

At the 100 km service-life case, well toward the generous end of the sensitivity range, the conventional ramp-and-sled method requires roughly twice the documented cedar supply for two timber components out of at least six. The documented benchmark covers less than half of the implied demand. Every assumption in the calculation is stacked in the model's favor.

6. THE SLED AS A CONSUMABLE

Fracture ceiling — new in v5.3

WORKBOOK SHEET: Fracture_Ceiling

The headline above wears the runners down by distance. But a three-ton block is not lowered onto the sled — it is dropped onto it, then the sled is snatched from rest against static friction on every start. That is impact and shock loading, not gradual abrasion, and wood under repeated shock **fractures**; it does not slowly thin. Fracture runs on a different clock than wear: the sled body fails on load-and-snatch cycles, not on kilometers.

This section runs that second clock. Two principles keep it from double-counting:

- **Runners are kept, not scrapped.** Like good tires moved off a dead car, runners are salvaged onto the next body and stay on their own slower wear clock — so they are counted once, on the anchored abrasion line, not re-counted with each body.
- **Strips are padding; the deck is structure.** The wear strips sit in the sled path and are consumed fast; the planking beneath is the structural deck, replaced half as often.

ASSUMPTIONS FOR THIS EXTENSION

Input	Value	Status / basis
Blocks per sled-body life	25	JUDGMENT — the single unmeasured variable here
Planking life vs wear strips	2× longer	Structural deck replaced half as often as padding
Sled-body volume (full sled – runners)	0.46 m³	0.69 m³ full sled – 0.23 m³ runner pair
Bollard standing inventory	300 m³	Allowance carried from omitted ledger
Waste & rejection	15%	Combined cut-loss + grade rejection; gross = net ÷ 0.85
Sled bodies required	92,000	2,300,000 blocks ÷ 25

COMPONENT DEMAND

Component	Demand (m³)	× benchmark	Clock
Runners (abrasion — salvaged, anchored)	5,605	2.04×	Distance — the bulletproof floor
Sled bodies (fracture)	42,437	15.46×	Cycles — rests on the 25-block call
Wear strips (padding)	282	0.10×	Fast wear
Ramp planking (structure)	1,071	0.39×	Half the strip frequency
Bollards (planking rate)	1,350	0.49×	—
Net subtotal (finished)	50,746	18.49×	
With 15% waste & rejection	59,701	21.75×	≈ 830 extra ships

SENSITIVITY — BLOCKS PER SLED-BODY LIFE

Blocks per body	× benchmark	Reference point
25	21.7×	Headline input for this section

Blocks per body	× benchmark	Reference point
50	12.7×	
100	8.1×	
250	5.4×	
500	4.5×	
1,000	4.0×	
2,000	3.8×	Very generous — body lasts ~2,000 blocks

Only the sled-body life is varied; runners, strips, planking, bollards, and waste are held constant.

READ THIS HONESTLY

The 21.75× figure rests on a single judgment: that a dropped-and-dragged sled body survives about 25 blocks before fracturing. That number has no direct bench test, and an opponent can dispute it without touching any other arithmetic. It is reported here because the physics behind it — impact and snatch loading fracture wood — is sound, and because even at a very generous 2,000-block body life the demand is still ~3.8× the documented supply. The defensible claim remains the anchored ~2.1× core; the fracture ceiling shows which direction every realistic correction pushes.

7. OMITTED TIMBER

What is NOT in the headline

WORKBOOK SHEET: Omitted_Timber

The headline includes only runners and track-strip planking. The categories below are excluded from it; each would increase demand and widen the gap. Mid-case rough volumes are shown for scale.

Category	Mid-case (m³)	Why the method needs it
Sled bodies beyond runners	840	Crossmembers, tow beam, deck, lash points
Posts, bollards, fairleads	300	Hauling control and corner geometry (2026 paper)
Heavy-lift sleepers / bedding	400	50–80 t granite block handling
Royal / funerary boats	300	Khufu boat alone $\approx$ 38 t cedar; competes for supply
Working cargo-boat fleet (conditional)	1,500	If logistics required cedar-planked craft
Conversion / cut loss (G)	1,472	Imported baulks → finished members, unavoidable
Structural-grade rejection (H)	3,963	Knots, sweep, twist, checks reduce usable yield

Adding the documented mid-case categories to the anchored headline pushes demand from 2.14× to roughly 4.8× the benchmark, before the fracture ceiling. The full low/mid/high stacked sensitivity is in the Faithful Export.

8. SOURCES

Every driving input traces to a source

WORKBOOK SHEET: Source_Ledger

Source	Use	Reference
2026 ramp-and-sled paper (npj Heritage Science)	Ramp haul, ramp geometry, IER parameters	doi.org/10.1038/s40494-026-02405-x
Ward, Seafaring in Ancient Egypt	40-ship Snefru cedar benchmark; Lebanon source	bibalex.org — Cheryl Ward
Palermo Stone	Documented cedar import under Snefru	Old Kingdom annalistic source
ASTM D2394	Wood abrasion — comparative benchmark only	ASTM standard
USDA / FPL wood wear data	Measured thickness-loss method	fpl.fs.usda.gov
USDA Wood Handbook / WWPA grading	Structural-grade yield: defects, grading	research.fs.usda.gov; wwpa.org
Brown / Joukowsky — Khufu barge	Cedar proxy for royal boats (~38 t)	Brown University archives
Great Pyramid block dimensions	1.27 × 1.27 × 0.71 m average	Petrie-derived measurements

Source policy. Every input that drives the calculation traces to a published source or to standard construction-engineering practice. Working values (haul averages, planked share, plank wear-rate fraction, sled-body life) are flagged as assumptions and sensitivity-tested. No input is taken from speculation or alternative-history sources.

9. HOW TO VERIFY THIS WORK

The calculation chain

Every number here was sourced, derived by arithmetic from sourced values, or flagged as an assumption. The anchored chain is:

1. Block mass = dimensions × density.
2. Runner thickness = minimum structural + 50% sacrificial.
3. Allowable wear depth = new thickness – minimum thickness.
4. Runner sets = total loaded haul ÷ service life × load factor.
5. Runner cedar demand = runner sets × replacement pair volume.
6. Plank cycles = cumulative track wear ÷ allowable plank wear.
7. Plank cedar demand = cycles × track-strip replacement inventory.
8. Combined demand = runner demand + plank demand.
9. Shortfall = combined demand – documented benchmark.
10. Demand multiple = combined demand ÷ documented benchmark.

Take the inputs from Section 2, run the chain, and the Section 5 numbers should fall out. The fracture-ceiling extension (Section 6) adds a parallel cycle clock for the sled body; its one judgment input is stated in the open and sensitivity-tested.

CLOSING FINDING

The conventional ramp-and-sled method, as published, does not close on its own material requirements. The documented cedar supply covers less than half of what the method needs for two timber elements out of at least six — and once the sled is counted as the consumable it physically is, the gap widens by an order of magnitude. If the math is wrong, show us where.